

Nishit Kumar

☎ +91-9060363204 | ✉ nishitkumaroll12@gmail.com | 🔗 LinkedIn | 🐙 GitHub | 📁 Portfolio

EDUCATION

National Institute of Technology Karnataka, Surathkal
B.Tech, Mining Engineering — Minor: ECE — CGPA: 8.45/10

Mangaluru, India
2023 – 2027

EXPERIENCE

Innovexis (*AI Engineer Intern*)

May – July

- Built an enterprise RAG chatbot with a LangGraph-orchestrated pipeline featuring hybrid retrieval (dense embeddings + BM25), cross-encoder re-ranking, and LLM-based query decomposition, improving retrieval accuracy and reducing irrelevant results.
- Implemented permission-aware access control, source-attributed responses, and LangSmith observability for production readiness, and validated answer quality using a custom RAGAS evaluation suite across 50+ QA pairs, improving answer faithfulness from 70% to 90%.

PROJECTS

Smart Shopping Agent | [GitHub](#) | *Python, LangGraph, LangChain, Groq, SerpAPI, Pydantic.*

- Built an agentic AI shopping assistant using LangGraph and Groq Llama 3.3 70B that classifies user intent, clarifies missing product requirements, and delivers ranked purchase recommendations instead of raw search results.
- Architected an 8 node LangGraph state machine with conditional routing across message classification, intent detection, query parsing, product search, deal validation, and price risk analysis, separating deterministic Python logic from LLM reasoning for reproducible recommendations.
- Implemented a deterministic pricing risk engine using Jaccard similarity peer grouping and median absolute deviation statistics to flag suspicious discounts, alongside a weighted recommendation scoring system combining requirement match, rating, reviews, and price value through SerpAPI Google Shopping integration.
- Engineered multi turn conversational memory with unique product ID tracking and streaming responses in a Streamlit interface, enabling follow up queries on previously shown products and reducing redundant LLM calls through cached client initialization.

Consumer Complaint Classifier | [GitHub](#) | *Python, TensorFlow, Keras, GloVe, Scikit-learn, ReportLab.*

- Built an end-to-end NLP deep learning system that classifies consumer financial complaints into 8 categories (credit cards, mortgages, debt collection, etc.) using BiLSTM, LSTM, and CNN architectures, achieving up to 83.6% accuracy and 0.82 macro F1-score with domain-fine-tuned GloVe embeddings.
- Architected a modular, leakage-free training pipeline (data ingestion → transformation → training → calibration → evaluation) with stratified 70/10/20 train/val/test splits, class-weighted loss for imbalance handling, and temperature-scaled confidence calibration.
- Deployed a production-style Streamlit application with occlusion-based token-importance explainability (XAI), head+tail truncation for long-text handling, and automated PDF report generation with input sanitization.

TECHNICAL SKILLS

Languages: C++, Python (OOPs), SQL

Libraries: NumPy, Pandas, Scikit-learn, TensorFlow, Keras, Matplotlib, Seaborn

NLP & GenAI: LangChain, LangGraph, Hugging Face Transformers, Text Embeddings, Semantic Search, RAG, SentenceTransformers, Groq API

Tools: Git, GitHub, Streamlit, Jupyter Notebook, VS Code, Excel, Tableau

Concepts: Machine Learning, Deep Learning, NLP, EDA, Data Cleaning, Statistical Analysis, Hypothesis Testing, A/B Testing, Feature Engineering

EXTRACURRICULAR

- Marketing Coordinator, Incident – NITK:** Led outreach efforts to 10+ potential sponsors, crafted sponsorship proposals, and collaborated with the marketing team to strengthen industry engagement for the fest.
- Class Representative (CR):** Serving as the Class Representative, acting as the primary liaison between students and faculty to coordinate academic schedules and address class-wide concerns.

ACHIEVEMENTS

- Solved 400+ Data Structures and Algorithms problems** on platforms like **LeetCode** and **GeeksforGeeks**, showcasing strong problem-solving and logical thinking skills.
- Academic Achievement:** Top 3% in JEE Adv/Mains (1M+ candidates).
- Awarded the Groww Scholarship:** a merit-based scholarship selected on CGPA performance — as one of the top-ranked students at NITK Surathkal.